ECE 313: Probability with Engineering Applications

2025 Fall   Instructors: Piao Chen & Xu Chen

Homework 9

Name: ____________________
Student ID: ____________________

Due Date: December 12 23:59, 2025

Problem 1. Let X, Y, and Z be independent Uniform(0,1) random variables.

(a) Find $P\left(\frac{X}{Y} \leq t\right)$ and $P(XY \leq t)$.

(b) Find $P\left(\frac{XY}{Z} \leq t\right)$.

Problem 2. Let X and Y be two independent random variables, where X has an $N(2,5)$ distribution and Y has an $N(5,9)$ distribution. Define $Z = 3X - 2Y + 1$.

(a) Compute $E(Z)$ and $\mathrm{Var}(Z)$.

(b) What is the distribution of Z?

(c) Compute $P(Z \leq 6)$.

Problem 3. Suppose X and Y have the joint pdf

$$f_{X,Y}(u,v) = \begin{cases} 2v, & 0 \le u \le 1,\ 0 \le v \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Are X and Y independent? Justify your answer.

(b) Find the pdf of $S = X + Y$.